

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	-2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

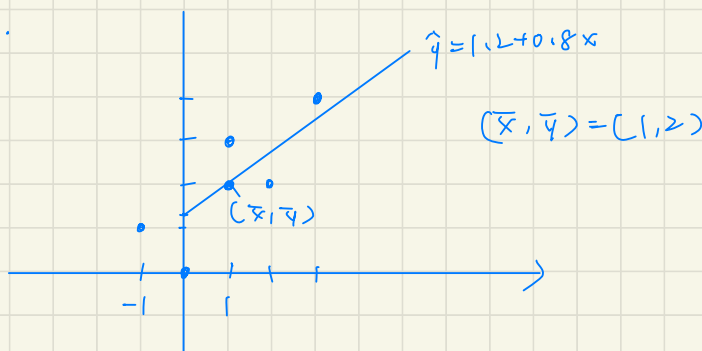
- a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?
b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.
c. Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.
d. Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y , $s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $s_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile, of x ?

$$\hat{y} = 1.2 + 0.8x$$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	2	-2	4	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

- e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.
f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.
g. Show that for these numerical values $\bar{y} = b_1 + b_2 \bar{x}$.
h. Show that for these numerical values $\bar{y} = \bar{y}$, where $\bar{y} = \sum \hat{y}_i / N$.
i. Compute $\hat{\sigma}^2$.
j. Compute $\text{var}(b_2 | x)$ and $\text{se}(b_2)$.

e. f.



g. $\bar{y} = 2$ $\bar{x} = 1$ 代入 $\hat{y} = 1.2 + 0.8x = 2$

h. $\bar{y} = \frac{\sum \hat{y}_i}{n} = \frac{10}{5} = 2 = \bar{y}$

i. $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

j. $\text{var}(b_2 | x)$ $\text{SE}(b_2)$
 $= \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$
 $= \sqrt{\text{var}(b_2 | x)} = \sqrt{0.12} = 0.3464$

$$y = b_1 + b_2 x$$

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b_2 = 8/10$$

$$b_1 = \bar{y} - b_2 \bar{x}$$

$$= 2 - 0.8 \times 1 = 1.2$$

$$\text{当 } x = 0 \text{ 时 } y = 1.2 = b_1$$

(b2)

每当 x 增加 1 单位,

y 预计增加 0.8 单位

C.

$$\sum x_i^2 = 15$$

$$N\bar{x}^2 = 5 \times 1^2 = 5$$

$$\sum (x_i - \bar{x})^2 = 10 = \sum x_i^2 - N\bar{x}^2$$

$$\sum x_i y_i = 18$$

$$N\bar{x}\bar{y} = 5 \times 1 \times 2 = 10$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8 = \sum x_i y_i - N\bar{x}\bar{y}$$

2.22 A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

- Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?
- Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?
- Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?
- Repeat part (c) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

a. $totalScore_i = 916.442 + 12.175 SMALL_i + e_i$
 当 $small = 0$ 时, $totalScore = 916.442$
 每当 $small$ 增加一单位时, $totalScore$ 上升 12.175 单位。
 $small$ 是显著影响 $totalScore$

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442    3.675 249.396  <2e-16 ***
small        12.175     5.369  2.268  0.0236 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608, Adjusted R-squared:  0.005323
F-statistic: 5.142 on 1 and 773 DF, p-value: 0.02363

```

b. $readScore = 432.665 + 6.924 small$
 当 $small = 0$ 时, $readScore = 432.665$
 每当 $small$ 增加一单位时, $readScore$ 上升 6.924 单位。

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665    1.488 290.760  <2e-16 ***
small        6.924     2.174  3.185  0.00151 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295, Adjusted R-squared:  0.01167
F-statistic: 10.14 on 1 and 773 DF, p-value: 0.001507

```

$mathScore = 483.117 + 5.251 small$
 当 $small = 0$ 时, $readScore = 483.117$
 每当 $small$ 增加一单位时, $readScore$ 上升 5.251 单位。
 $small$ 对 $MathScore$ 效果不够显著

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777    2.449 197.545  <2e-16 ***
small        5.251     3.578  1.467  0.143
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778, Adjusted R-squared:  0.001488
F-statistic: 2.153 on 1 and 773 DF, p-value: 0.1427

```

c. $totalScore_i = 916.442 + 4.306 aide$
 当 $small = 0$ 时, $totalScore = 916.442$
 每当 $aide$ 增加一单位时, $totalScore$ 上升 4.306 单位。
 老师的帮助并不显著

```

> summary(reg5)

Call:
lm(formula = readscore ~ small + aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-74.665 -21.536  -2.665  15.464 194.335

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665    1.487 291.006  <2e-16 ***
small        6.924     2.172  3.187  0.00147 **
aide         2.871     2.086  1.376  0.16902
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.18 on 1197 degrees of freedom
Multiple R-squared:  0.00845, Adjusted R-squared:  0.006794
F-statistic: 5.101 on 2 and 1197 DF, p-value: 0.006226

```

$readScore = 432.665 + 2.871 aide + 6.924 small$
 当 $small = 0$ 时, $totalScore = 432.665$
 每当 $aide$ 增加一单位时, $readScore$ 上升 2.871 单位。
 老师的帮助并不显著

```

> summary(reg6)

Call:
lm(formula = mathscore ~ small + aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-146.212 -34.777  -5.777  29.223 142.223

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777    2.391 202.308  <2e-16 ***
small        5.251     3.494  1.503  0.133
aide         1.435     3.356  0.428  0.669
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.54 on 1197 degrees of freedom
Multiple R-squared:  0.001985, Adjusted R-squared:  0.0003173
F-statistic: 1.19 on 2 and 1197 DF, p-value: 0.3045

```

$\text{mathscore} = 483.777 + 1.435 \text{aide} + 5.25 \text{small}$
 当 $\text{small} = 0$ 时, $\text{totalscore} = 483.777$
 每当 aide 增加一单位时, mathscore 上升 1.435 单位。
 aide from teacher isn't significant for math

```

> summary(lm7)

Call:
lm(formula = totalscore ~ small + aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-191.75  -50.75   -8.44   42.28  324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442      3.615  253.525  <2e-16 ***
small         12.175       5.282   2.305  0.0213 *
aide          1.436       5.073   0.283  0.3861
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

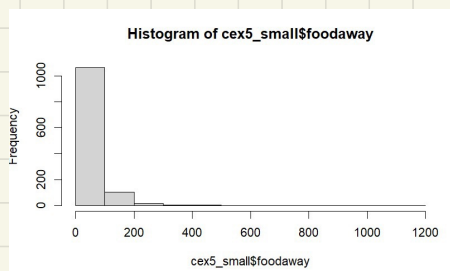
Residual standard error: 73.37 on 1197 degrees of freedom
Multiple R-squared:  0.004502, Adjusted R-squared:  0.002839
F-statistic: 2.707 on 2 and 1197 DF, p-value: 0.06717
  
```

mated coefficient of *DREC*.

2.25 Consumer expenditure data from 2013 are contained in the file *cex5_small*. [Note: *cex5* is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between \$1000 per month to \$20,000 per month. *FOODAWAY* is past quarter's food away from home expenditure per month per person, in dollars, and *INCOME* is household monthly income during past year, in \$100 units.

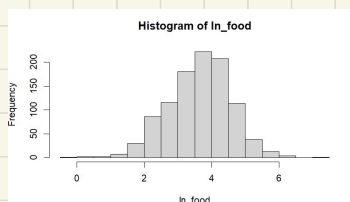
- Construct a histogram of *FOODAWAY* and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?
- What are the mean and median values of *FOODAWAY* for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?
- Construct a histogram of $\ln(\text{FOODAWAY})$ and its summary statistics. Explain why *FOODAWAY* and $\ln(\text{FOODAWAY})$ have different numbers of observations.
- Estimate the linear regression $\ln(\text{FOODAWAY}) = \beta_1 + \beta_2 \text{INCOME} + e$. Interpret the estimated slope.
- Plot $\ln(\text{FOODAWAY})$ against *INCOME*, and include the fitted line from part (d).
- Calculate the least squares residuals from the estimation in part (d). Plot them vs. *INCOME*. Do you find any unusual patterns, or do they seem completely random?

a. mean = 49.27085
 median = 32.555
 25th percentile = 12.04
 75th percentile = 67.5



b. advanced : mean = 73.15494
 median = 48.15
 college : mean = 48.59718
 median = 36.11
 none : mean = 39.01017
 median = 26.02

C. 观测到不同的原因: $\log(c)$ doesn't exist.



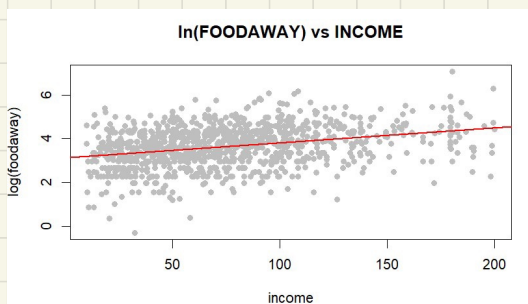
d. $\ln(\text{Foodaway}) = 3.1293 + 0.0069 \text{Income}$.

当 $\text{income} = 0$ 时, $\ln(\text{foodaway}) = 3.1293$

每当 income 增加一单位时, $\ln(\text{foodaway}) \uparrow 0.0069$ 单位.

Income 是显著变量

e.



f. 食物支出的波动并没有因为所得不同而不同

